

Orion: Spatiotemporal Neural Networks with Self-Validating Directed Graphs for Traffic and Epidemic Forecasting

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Abstract

Spatiotemporal forecasting is fundamental to intelligent transportation and epidemic prediction, yet existing methods face challenges in modeling complex inter-node dependencies and multi-scale temporal dynamics. A key limitation lies in their reliance on symmetric correlations, which fail to capture how perturbations propagate directionally through real-world networks. To address this gap, we propose Orion, a framework that learns directed dependency structures grounded in Granger causality principles and validates them through prediction-based consistency testing—a mechanism notably absent in prior spatiotemporal graph learning methods that provides a rigorous training signal beyond pure correlation by testing whether learned edge weights remain consistent with their predictive effects. Building on this foundation, our framework further integrates: (1) the TE-CausGAT module, which estimates time-varying directed dependencies considering both lagged and instantaneous effects and integrates them with prior spatial structures through prior-guided fusion; (2) the Belt Block architecture for efficient parallel multi-scale temporal modeling across hourly, daily, and weekly patterns; (3) a three-stage progressive training strategy that transitions from feature learning to directed structure learning to end-to-end optimization. Empirically, Orion achieves state-of-the-art performance on multiple real-world benchmarks: in traffic forecasting, it improves MAE by 6.60% on METR-LA and 8.65% on PEMS-08, with notable short-horizon gains on PEMS-04 though long-horizon performance remains competitive rather than dominant; in epidemic forecasting, it improves MAE by 22.70% on CA and 37.73% on TX. DREAM-3 experiments confirm that learned directed structures align well with ground-truth regulatory networks, while case studies on METR-LA demonstrate that discovered dependencies correspond to known traffic propagation patterns.

CCS Concepts

• Information systems → Data mining; Spatial-temporal systems; • Applied computing → Forecasting.

Keywords

spatiotemporal forecasting, Granger causal graph learning, graph attention networks, intervention-based validation, multi-scale temporal modeling, multi-head attention

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1 Introduction

Spatiotemporal forecasting underpins the modeling of complex dynamical systems across intelligent transportation and infectious disease prediction. What most existing methods overlook is that real-world spatiotemporal systems evolve through directed dependency structures rather than merely symmetric correlations.

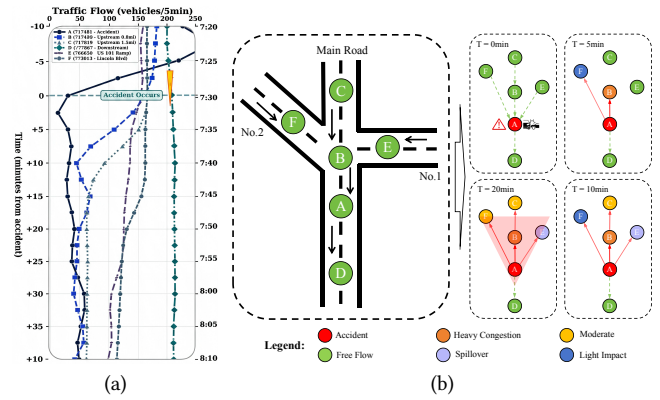


Figure 1: Directed dependency propagation in METR-LA traffic networks. (a) Traffic flow dynamics showing six nodes before/after accident at $T=0$; node labels (A-F) correspond to sensor IDs in legend. (b) Spatial snapshots: left shows road topology; right shows temporal evolution at $T=0, 5, 10, 20$ min. Node colors indicate congestion severity (green→red). Green dashed arrows: traffic flow direction; red solid arrows: influence direction discovered by our model.

To illustrate, we examine a highway interchange scenario during morning rush hour (Figure 1). Traffic flows along the main road $C \rightarrow B \rightarrow A \rightarrow D$, with branch roads connecting nodes E and F. At $T=0$ min, an accident at Node A triggers *asymmetric* influence propagation: A affects upstream B and branch F, but not downstream D. By $T=5$ min, B shows heavy congestion while E and F remain unaffected. By $T=10$ min, congestion propagates further to C ($B \rightarrow C$) with E lightly impacted. By $T=20$ min, the directed chain $A \rightarrow B \rightarrow C$ is fully established, F is more congested than E due to its more direct connection to A, and D remains in free-flow throughout. Discovering such directed dependencies—where influence propagates asymmetrically with varying intensities—is precisely the goal of Orion, enabling more accurate forecasting by leveraging these asymmetric predictive relationships.

Existing spatiotemporal forecasting methods have evolved from generic architectures to specialized designs. Early studies applied general deep learning architectures to spatiotemporal data: RNNs [34] and Transformers [39] model temporal dependencies, while GCN [24] and TCN [21] based on convolutions, and attention-based GAT [40], capture spatial patterns in graph-structured data. However, these architectures lack spatiotemporal coupling and struggle to capture complex spatial-temporal interactions simultaneously. The second category designs coupled architectures for spatiotemporal forecasting: DCRNN [25] integrates diffusion convolution with GRU, GWNet [42] introduces adaptive adjacency matrix learning, and AGCRN [3] and DGCRN [23] achieve spatiotemporal coupling through message passing. However, these methods typically assume static graph structures and treat inter-node dependencies as symmetric relationships, ignoring dependency directionality inherent in real-world spatiotemporal systems. The third category explores causal-inspired modeling: CaST [43] applies structural causal models to address confounding, while TESTAM [22] adaptively selects spatial modeling strategies. However, these methods either rely on strong assumptions or lack explicit directed dependency validation.

Infectious disease forecasting suffers from the same directional bias. Pathogens spread along directional human mobility networks, so transit-hub counties seed peripheral ones far more readily than the reverse, with reporting lags and shared policies further breaking pairwise symmetry [10]. Yet conventional spatiotemporal models cast county-level adjacency as an undirected graph with symmetric correlations, conflating “A drives B” with “A and B share a common driver” [32, 47], thereby inflating edges between large neighbouring populations and missing long-range, one-way transmission pathways. Modelling the directionality—not merely the magnitude—of such dependencies is thus as critical for epidemic as for traffic forecasting, motivating a unified application of Orion across both domains.

Despite advances in specific scenarios, these methods exhibit four limitations: (1) neglecting dependency directionality, unable to distinguish influence sources from targets; (2) static dependency assumptions hindering dynamic structural adaptation; (3) single-scale temporal modeling struggling to capture multi-period features; (4) no existing method validates whether learned spatiotemporal graph structures are consistent with their predictive effects. We propose Orion to address these issues, with the following contributions:

- We introduce a prediction-based consistency validation mechanism that, to our knowledge, is the first to validate learned spatiotemporal dependency structures by testing whether edge weights predict the actual sensitivity of downstream nodes to upstream masking, transcending correlation-based graph learning by providing a self-consistency training signal that filters spurious edges.
- We design the TE-CausGAT module that estimates time-varying directed dependencies in the Granger sense, providing interpretable and validatable dependency structures whose primary value lies in structural discovery—confirmed by DREAM-3 alignment with ground-truth networks—and in growing importance at longer horizons. Prior-guided fusion integrates learned structures with prior spatial knowledge for stability.

- We introduce the Belt Block architecture combining TE-CausGAT with multi-head attention for parallel multi-scale temporal modeling, trained via a three-stage progressive strategy from feature learning to directed structure learning to joint optimization.
- Orion achieves state-of-the-art results across five benchmarks spanning traffic and epidemic domains, with DREAM-3 experiments confirming that learned structures align with ground-truth directed regulatory networks (AUC 0.6295 vs. 0.5915 for the best baseline).

Our code and supplementary materials are available at https://anonymous.4open.science/r/Orion_Supplementary_Material/

2 Preliminaries

Definition 1 (Spatiotemporal Forecasting). Given a spatial network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ with N nodes and adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, let $\mathbf{X}^{(t)} \in \mathbb{R}^{N \times F}$ denote node features at time t . The task is to learn $f: \mathbb{R}^{N \times F \times T} \times \mathcal{G} \rightarrow \mathbb{R}^{N \times H}$ mapping historical observations $\{\mathbf{X}^{(t-T+1)}, \dots, \mathbf{X}^{(t)}\}$ to future predictions $\{\hat{\mathbf{X}}^{(t+1)}, \dots, \hat{\mathbf{X}}^{(t+H)}\}$.

Definition 2 (Granger-inspired Directed Dependency Graph). Following Granger causality—where X Granger-causes Y if past X improves prediction of Y —we define a dynamic directed dependency graph $\mathcal{C} = \{\mathbf{C}^{(s)}\}_{s=1}^S$, where $\mathbf{C}^{(s)} \in [0, 1]^{N \times N}$ captures directed predictive dependencies for temporal segment s . Element $\mathbf{C}_{ij}^{(s)}$ quantifies predictive influence from v_i to v_j , exhibiting asymmetry ($\mathbf{C}_{ij}^{(s)} \neq \mathbf{C}_{ji}^{(s)}$) and acyclicity ($\text{tr}(\mathbf{e}^{\mathbf{C}^{(s)}}) - N = 0$) [46].

Definition 3 (Prediction-Based Consistency Validation). We validate learned dependencies through prediction-based testing: given edge weight $\mathbf{C}_{ij}$, we mask node v_i 's contribution and measure the resulting prediction change at node v_j . High consistency between learned weights and observed prediction changes indicates that the dependency structure captures genuine predictive relationships rather than spurious correlations.

3 Methodology

As shown in Figure 2, during Orion's forward propagation, the input data generates three period tensors $\mathbf{X}_h, \mathbf{X}_d, \mathbf{X}_w \in \mathbb{R}^{N \times F \times T}$ through the LLM-enhanced retrieval mechanism, which are then transformed into embedded representations $\mathbf{E} \in \mathbb{R}^{N \times d_{\text{model}} \times T}$ through embedding layers with fixed sinusoidal positional encodings [39] added element-wise to the embedded features and shared across the hourly, daily and weekly branches. Three parallel Belt Blocks independently process period data, sequentially performing TE-CausGAT spatial attention, multi-head temporal attention, and feedforward network operations, interspersed with residual connections [14] and layer normalization [2]. The fusion module receives branch outputs $\{\mathbf{H}_h, \mathbf{H}_d, \mathbf{H}_w\} \in \mathbb{R}^{N \times d_{\text{model}}}$, integrates three-period features through adaptive weighting and attention, and generates predictions $\mathbf{X}_{\text{output}} \in \mathbb{R}^{N \times H}$ through linear projection. Notably, Orion adopts three-stage training: the first stage learns feature representations, the second introduces directed dependency regularization with frozen embedding layers, and the third unfreezes parameters for end-to-end fine-tuning.

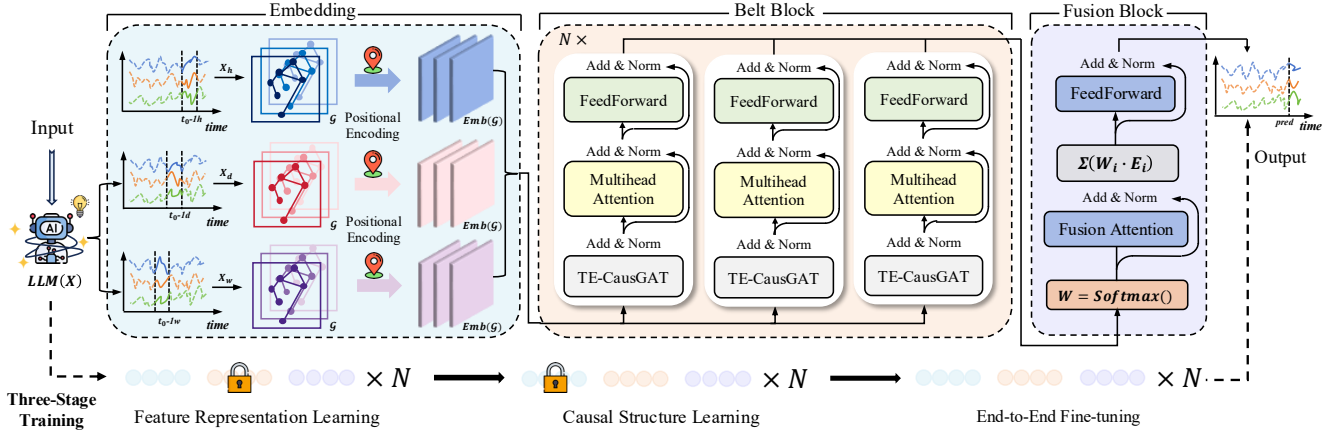


Figure 2: Overall architecture of the Orion framework. Three parallel Belt Blocks process hourly, daily, and weekly period data independently. Bottom: three-stage progressive training strategy.

3.1 Fine-tuned LLM-enhanced Data Retrieval

Traditional methods rely on fixed temporal windows to select historical reference data, which performs poorly when encountering non-periodic variations. For example, traffic patterns during holidays differ significantly from those on regular weekdays, and fixed retrieval of “the same time last week” may introduce irrelevant pattern noise. To overcome this limitation, we propose a semantic similarity-based intelligent retrieval mechanism. By fine-tuning the TinyLlama-1.1B model on 100,000 temporal samples from the PEMS03 dataset (details in Appendix C), the LLM learns to understand semantic characteristics of temporal patterns, thereby enabling retrieval of truly similar historical patterns across calendar boundaries. As shown in Figure 3, taking the daily cycle as an example, both the input sequence $X_h \in \mathbb{R}^{N \times F \times T}$ at current time t and the data from the same time periods over the past D days are encoded using a fine-tuned large language model, extracting the average of the last hidden layer as the semantic embedding $z_i \in \mathbb{R}^{d_{embed}}$. The similarity between the current sequence and candidate historical sequences is measured through a weighted combination of cosine similarity and Euclidean distance:

$$S_{sim}(i, j) = w_{cos} \cdot \frac{z_i^T z_j}{\|z_i\|_2 \cdot \|z_j\|_2} + w_{euc} \cdot \frac{1}{1 + \|z_i - z_j\|_2} \quad (1)$$

Each node selects its most similar historical cycle as X_d . It is important to emphasize that, with or without the LLM, all input data are embedded through the same fully connected layer with parameter dimensions $F \times d_{model}$, followed by positional encoding before being processed by the Belt Block. The role of the LLM is intelligent data selection rather than feature extraction. TinyLlama-1.1B-Chat-v1.0 is fine-tuned offline once on PEMS03 (see Appendix C) and then frozen throughout Orion’s training: the model is invoked exclusively in inference mode and all queries are performed without gradient tracking, so its parameters receive no gradient updates from Orion’s prediction loss and serve purely as an inference-time semantic retriever.

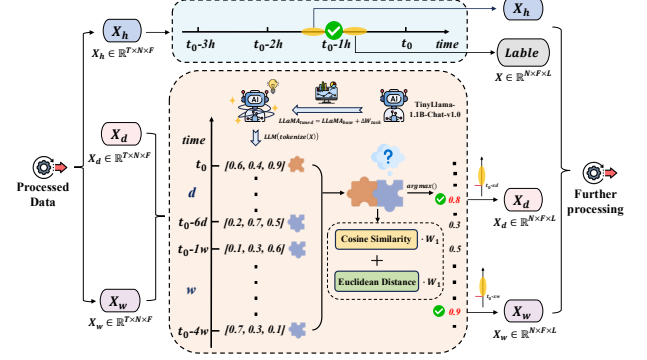


Figure 3: Fine-tuned LLM-enhanced data retrieval. X_h is directly extracted; X_d and X_w are selected via semantic similarity using fine-tuned TinyLlama-1.1B-Chat-v1.0.

3.2 Temporal Evolution Causal Graph Attention Network (TE-CausGAT)

As shown in Figure 4, TE-CausGAT is the core module for learning and validating directed dependency structures in Orion. We adopt the term “causal” in the Granger causality sense throughout this work: a variable X is said to Granger-cause Y if the past values of X contain information that helps predict Y beyond what is contained in past values of Y alone [11]. This predictive notion of causality is weaker than Pearl’s interventional causality but is identifiable from observational time series data without requiring controlled experiments. Our TE-CausGAT module learns such Granger-causal directed dependencies and validates them through prediction-based consistency testing.

3.2.1 Granger-inspired Directed Dependency Learning. The left pathway of Figure 4 chains multi-scale temporal convolution, segment-wise dependency estimation, prior-guided fusion, and dependency-modulated propagation, while the right pathway performs prediction-based validation during training to filter spurious edges. Convolutions on X_{input} yield multi-scale features through

fusion in TE-CausGAT. Next, Temporal segmentation divides the series into S segments to capture causal evolution. Each segment s uses an independent estimator for its causal structure. For segment s , causal relationship strengths are computed by learning source and target node transformation matrices through time-weighted feature aggregation combined with node embeddings:

$$C^{(s)} = \sigma(S^{(s)} \cdot (T^{(s)})^T + B) \quad (2)$$

Where $S^{(s)}, T^{(s)}$ are source and target transforms, B is a bias matrix, σ is the sigmoid function, and the self-loop mask removes diagonal elements. The prior-guided causal fusion mechanism integrates segment-specific causal matrices through attention weights (the prior graph structure does not explicitly contain causal relationships) and incorporates prior structures from the adjacency matrix:

$$C_{\text{fused}} = (1 - \alpha_{\text{prior}}) \cdot \sum_{s=1}^S w_s \cdot C^{(s)} + \alpha_{\text{prior}} \cdot C_{\text{prior}} \quad (3)$$

where w_s denotes the attention weight for segment s satisfying $\sum_{s=1}^S w_s = 1$, C_{prior} the prior adjacency, and α_{prior} balances discovered vs prior structures for stability. Subsequently, the dependency propagation simulator employs a GRU mechanism [8] to model the temporally dynamic propagation of perturbation effects:

$$h^{(t)} = \text{GRU}(C_{\text{fused}} \cdot X^{(t)}, h^{(t-1)}) \quad (4)$$

where $X^{(t)} \in \mathbb{R}^{N \times d_{\text{model}}}$ is the node features at time t , $h^{(t)} \in \mathbb{R}^{N \times d_{\text{model}}}$ is the hidden state, and the matrix product $C_{\text{fused}} \cdot X^{(t)}$ modulates information flow between nodes. The causal matrix modulates information flow between nodes and simulates the temporal propagation effects of node changes. To ensure the learned causal graph maintains acyclicity, we employ computationally efficient polynomial approximations instead of the theoretically exact but computationally expensive matrix exponential constraint $\text{tr}(e^{C^{(s)}}) - N = 0$. Specifically, we penalize the trace of powers of the normalized causal matrix: $\mathcal{L}_{\text{DAG}} = \lambda_2 \cdot \text{tr}(\hat{C}^2) + \lambda_3 \cdot \text{tr}(\hat{C}^3)$, where $\hat{C} = C_{\text{fused}} / \|C_{\text{fused}}\|_{\infty}$, $\lambda_2 = 0.1$ and $\lambda_3 = 0.01$. This exploits $\text{tr}(C^k)$ counting k -cycles, suppressing cycles while remaining differentiable and efficient. Although the polynomial approximation cannot theoretically guarantee complete acyclicity, the causal graphs discovered by Orion strictly satisfy the DAG property in subsequent experimental learning (see our supplementary appendix, Section S.2).

3.2.2 Intervention-inspired Consistency Validation. The core feature of TE-CausGAT is an intervention-inspired validation mechanism. Figure 5 illustrates how this mechanism tests whether learned edge weights predict intervention responses. When intervening on node i , the effect on downstream node j can be expressed under linear approximation as $\Delta \hat{y}_j \approx C_{ij} \cdot \Delta X_i + \sum_{k \neq i} C_{kj} \cdot \Delta X_k$. We implement this mechanism through four steps: randomly selecting K intervention nodes for unbiased discovery; generating intervention values $v = \tanh(g(c))$ constrained to $[-1, 1]$; implementing the masking operation by setting node values to v_i and cutting incoming edges, then propagating through GRU to simulate dynamics; finally evaluating consistency between causal strength and observed effects through validity score, which is defined as:

$$V_{i \rightarrow j} = 1 - \left| C_{ij} - \frac{\Delta \hat{y}_j}{\max_k (\Delta \hat{y}_k) + \epsilon_{\text{stab}}} \right| \quad (5)$$

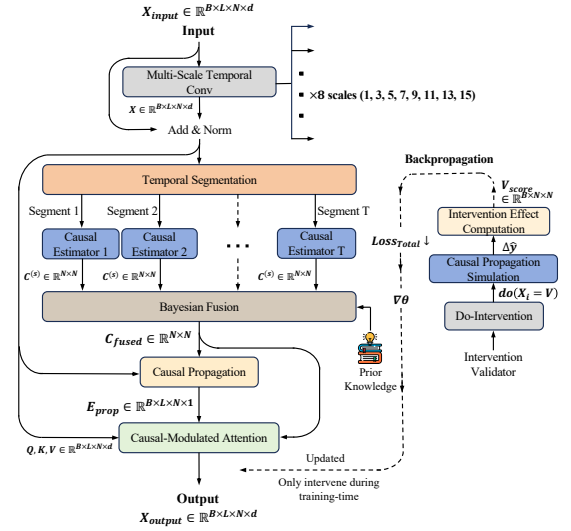


Figure 4: Architecture of TE-CausGAT. Left: forward pathway from multi-scale convolution through directed dependency estimation to dependency-modulated attention, operating on the fused causal matrix C_{fused} and node features X . Right: training-only validation pathway taking intervention-perturbed features X' produced by $\text{do}(\cdot)$ (Section 3.2.2), where comparing the downstream prediction change against learned edge weights yields the validity score that filters spurious edges.

This formulation is based on the causal invariance assumption: if systematically removing X_a from $Y = f(X_a, X_b, X_c, X_d)$ yields significant change in Y , then X_a carries predictively relevant influence on Y . We use $\mathcal{L}_{\text{validity}}$ to promote high validity on causal edges, effectively filtering correlations that fail intervention tests. where the normalized empirical effect $\frac{\Delta \hat{y}_j}{\max_k (\Delta \hat{y}_k) + \epsilon_{\text{stab}}}$ quantitatively estimates the relative causal influence, the $C_{ij} \in [0, 1]$ represents causal strength from node i to node j , $\Delta \hat{y}_j = |\hat{y}_j^{\text{post}} - \hat{y}_j^{\text{pre}}|$ is the prediction change for node j , and $\epsilon_{\text{stab}} = 10^{-8}$ prevents division by zero. The do-operation is an internal computationally simulated intervention used for validation rather than a physical intervention in the real world. A high validity score indicates consistency between the learned causal strength and the intervention effect, helping the model capture predictive dependencies, while a low score clearly reveals spurious correlations. We emphasize that this mechanism validates *predictive relevance* rather than interventional causality in the Pearl sense. It provides a robust training signal beyond pure correlation, thereby enabling Orion to filter spurious correlations and learn directed dependency structures that are consistent with their predictive effects—a desirable property we term *prediction-based consistency*.

3.2.3 Causal-Modulated Attention. As illustrated by the Fusion Block in Figure 2, the validated and fused causal matrix C_{fused} is injected into multi-head attention through two complementary pathways. The validated causal matrix enhances multi-head attention. Q, K, V are projected from multi-scale temporal features. This

mechanism operates through two complementary pathways that synergistically integrate causal knowledge: (1) dynamically scaling the value matrix based on causal propagation effects, where a three-layer MLP generates position-specific modulation strengths $m(\mathbf{X}, \mathbf{E})$ to amplify information on strong causal paths while suppressing weak paths; (2) directly enhancing attention scores using the fused causal matrix. The complete formulation is:

$$\text{Output} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \alpha_{\text{gate}} \lambda_{\text{scale}} \mathbf{C}_{\text{fused}} \right) \cdot \left[\mathbf{V} \odot (1 + \alpha_{\text{gate}} m(\mathbf{X}, \mathbf{E}_{\text{prop}}) \mathbf{E}_{\text{prop}}) \right]. \quad (6)$$

where α_{gate} is a learnable gate, λ_{scale} is a fixed scaling coefficient, $\mathbf{C}_{\text{fused}}$ represents the fused causal matrix, $\mathbf{E}_{\text{prop}}$ denotes propagation effects, and $m(\cdot)$ is a three-layer MLP modulation function. This design dynamically directs attention along validated causal paths and adaptively adjusts their strength.

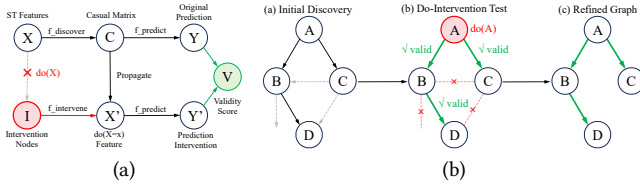


Figure 5: Prediction-based validation mechanism. (a) Masking a node’s contribution generates a prediction change; the validity score measures its consistency with the learned edge weight. (b) Candidate edges are tested and spurious correlations (red crosses) are filtered. Here $\text{do}(\cdot)$ denotes the masking intervention defined in Section 3.2.2 (zeroing the node’s input and cutting its incoming edges). The two pathways differ in their inputs: the left pathway feeds $\mathbf{C}_{\text{fused}}$ and node features into the predictor, while the right pathway feeds the perturbed features $\mathbf{X}'$ to measure the downstream prediction change; comparing the learned edge weights with this change yields the validity score in Eq. 5.

3.3 Belt Block Architecture and Multi-Period Attention Fusion

The Belt Block serves as our fundamental unit, sequentially applying spatial causal attention, temporal attention, and feedforward transformation, augmented with residual connections. We instantiate three parallel Belt Block branches corresponding to Hour, Day, and Week periodicities, where each branch processes period-specific data selected through our LLM-enhanced retrieval mechanism. These branches independently capture distinct periodic patterns while maintaining computational efficiency via TE-CausGAT parameter sharing.

3.4 Three-Stage Progressive Training Strategy

Following curriculum learning [4], we adopt a three-stage progressive training strategy. It moves from feature learning to causal discovery to joint optimization, reducing local minima in complex

end-to-end training (weight scheduling in our supplementary appendix, Section S.3.3).

Stage 1: Feature Representation Learning (Correlation-Based Learning). Freeze all causal-related modules and focus on spatiotemporal feature representation learning:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{pred}} = \text{MAE}(\mathbf{Y}, \hat{\mathbf{Y}}) \quad (7)$$

This ensures embedding layers, multi-scale convolutions, and attention mechanisms develop stable spatiotemporal representations.

Stage 2: Directed Structure Learning (Dependency Hypothesis Formation). Freeze embedding modules and concentrate learning capacity on TE-CausGAT components:

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{causal}} \mathcal{L}_{\text{causal_total}}, \quad \mathcal{L}_{\text{causal_total}} = \sum_i w_i \mathcal{L}_i. \quad (8)$$

Concretely, in this stage we set `requires_grad=False` for every parameter whose module name contains `embedding_process` (the shared feature feed-forward block of Section 3.1) and leave all other parameters trainable, so that the multi-scale temporal convolutions, the segment estimators $\mathbf{S}^{(s)}$, $\mathbf{T}^{(s)}$, $\mathbf{B}$, the segment attention weights $\mathbf{w}_s$, the prior-guided fusion gate α_{prior} , the GRU-based propagation simulator, the intervention validator and the causal-modulated attention all receive gradients. The learnable TE-CausGAT parameters are not re-initialized at Stage 2; they are inherited from the Stage 1 checkpoint and were originally instantiated using PyTorch defaults—Kaiming-uniform initialization for every `nn.Linear` in $\mathbf{S}^{(s)}$, $\mathbf{T}^{(s)}$ and the time-delay/intervention/validity MLPs, $\mathcal{N}(0, 1)$ for the segment, node and time embeddings, $\mathcal{N}(0, 0.01^2)$ for the bias matrix $\mathbf{B}$, the prior adjacency matrix for $\mathbf{C}_{\text{prior}}$, 0.3 for the prior confidence and 0 for the causal gate α_{gate} .

Stage 3: End-to-End Optimization (Intervention-based Refinement). Unfreeze all parameters for joint optimization of feature learning and directed structure learning:

$$\mathcal{L}_{\text{stage3}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{causal}} \cdot \mathcal{L}_{\text{causal_total}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{regularization}} \quad (9)$$

4 Experiments

Datasets. We evaluate Orion on five spatiotemporal datasets: for traffic forecasting, we employ PEMS04, PEMS08 [6], and METR-LA for traffic flow prediction; we utilize CA-COVID and TX-COVID datasets [10] for hospitalization prediction (details in Appendix A). **Baselines.** We compare against 13 methods: statistical models (HA, VAR), RNN-based models (DCRNN, AGCRN, DGCRN), CNN or attention-based models (STGCN, GWNet, ASTGCN, MTGNN [41], GMAN [45], STAEformer [26]), and causal-aware models (CaST, TESTAM).

Implementation Details. The adjacency matrix is constructed using a Gaussian kernel with σ_{kernel} set to the median of non-zero distances. Datasets are partitioned 6:2:2 for traffic (7:1:2 for METR-LA) and 8:1:1 for epidemic data. LLM retrieval is an optional module; to ensure fair comparison, main results (Table 1) use fixed-period retrieval, while LLM retrieval is evaluated independently (Table 4) to demonstrate semantic retrieval potential. Evaluation metrics are MAE, RMSE, MAPE (with threshold 5 for near-zeros), and L1 loss. We use Adam [19] for 100 epochs. All baselines and Orion adopt the same setup for fair comparison (see Appendix B for details).

Table 1: Traffic forecasting performance comparison with 12-step prediction horizon. MAPE values in %. Red bold: best performance, underline: second best.

Dataset	Method	Horizon 3			Horizon 6			Horizon 12			Average		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
METR-LA	HA	4.79	10.00	11.70	5.47	11.45	13.50	6.99	13.89	17.54	5.64	11.74	13.85
	VAR	4.42	7.80	13.00	5.41	9.13	12.70	6.52	10.11	15.80	5.23	9.09	13.26
	DCRNN	2.78	5.36	7.30	3.16	6.43	8.80	3.61	7.57	10.50	3.19	6.35	8.76
	AGCRN	2.88	5.56	7.70	3.24	6.57	9.01	3.69	7.55	10.46	3.24	6.40	8.83
	DGCRN	2.63	5.00	6.64	3.00	6.04	8.03	3.45	7.18	9.74	3.00	6.15	7.81
	STGCN	2.88	5.74	7.61	3.47	7.24	9.56	4.59	9.40	12.69	3.56	7.16	9.58
	GWNet	2.69	5.15	6.90	3.07	6.22	8.37	3.53	7.37	10.01	3.06	6.19	8.37
	ASTGCN	4.84	9.25	9.19	5.41	10.59	10.11	6.48	12.49	11.61	5.38	10.90	10.38
	MTGNN	2.68	5.17	6.86	3.04	6.16	8.17	3.48	7.22	9.85	2.98	5.79	7.92
	GMAN	2.80	5.55	7.41	3.12	6.49	8.73	3.44	7.35	10.07	3.07	6.39	8.57
	STAEformer	2.65	5.11	6.85	2.97	6.00	8.13	3.34	7.02	9.70	<u>2.85</u>	<u>5.93</u>	8.25
	CaST	2.71	5.26	7.14	3.08	6.29	8.52	3.42	7.21	10.03	3.01	6.04	8.44
	TESTAM	<u>2.54</u>	<u>4.93</u>	<u>6.42</u>	<u>2.96</u>	<u>6.04</u>	<u>7.92</u>	<u>3.36</u>	<u>7.09</u>	<u>9.67</u>	2.91	5.96	<u>7.76</u>
	Orion	1.83	2.87	3.88	2.70	5.22	5.85	3.73	7.49	8.14	2.66	5.40	5.78
PEMS04	HA	28.92	42.69	20.31	33.73	49.37	24.01	46.97	67.43	35.11	36.49	52.93	25.46
	VAR	21.94	34.30	16.42	23.72	36.58	18.02	26.76	40.28	20.94	23.51	36.39	17.85
	DCRNN	18.51	29.59	12.69	19.63	31.35	13.43	21.65	34.17	15.01	19.69	31.41	13.52
	AGCRN	18.52	29.79	12.31	19.45	31.45	12.82	20.64	33.31	13.74	19.36	31.28	12.81
	DGCRN	18.27	28.97	12.36	19.39	30.86	13.42	21.09	33.59	14.94	19.39	<u>29.18</u>	13.46
	STGCN	18.74	29.84	14.42	19.64	31.34	13.27	21.12	33.53	14.22	19.63	31.32	13.32
	GWNet	18.02	28.81	13.66	18.98	30.31	14.25	20.55	32.52	15.43	18.99	30.30	14.28
	ASTGCN	19.35	30.58	13.41	21.24	33.42	14.78	25.08	38.95	18.35	21.37	34.04	15.21
	MTGNN	18.65	30.13	13.32	19.48	32.02	14.08	20.96	34.66	14.96	19.50	32.00	14.04
	GMAN	18.26	29.34	12.65	18.80	30.84	13.24	20.00	31.31	13.39	18.82	30.92	13.20
	STAEformer	17.92	29.18	12.26	<u>18.74</u>	30.73	12.81	20.18	33.01	13.72	18.95	30.97	12.93
	CaST	18.36	29.51	12.48	19.21	31.18	13.15	20.72	33.76	14.28	19.13	28.76	12.76
	TESTAM	<u>17.79</u>	<u>29.01</u>	<u>12.17</u>	18.62	<u>30.56</u>	<u>12.72</u>	<u>20.03</u>	<u>32.84</u>	<u>13.61</u>	18.81	30.80	<u>12.83</u>
	Orion	16.22	24.52	11.73	18.86	28.65	12.63	24.53	36.74	15.20	19.04	29.31	12.52
PEMS08	HA	23.52	34.96	14.72	27.67	40.89	17.37	39.28	56.74	25.17	28.64	42.27	18.21
	VAR	19.52	29.73	12.54	22.25	33.30	14.23	26.17	38.97	17.32	22.07	31.02	14.04
	DCRNN	14.18	22.18	9.33	15.26	24.24	9.92	17.72	27.12	11.15	15.28	24.26	9.98
	AGCRN	14.52	22.86	9.35	15.67	24.99	10.35	17.50	27.92	11.73	15.66	24.98	10.18
	DGCRN	13.89	22.07	9.19	14.92	23.99	9.85	16.73	26.88	10.84	15.01	24.62	9.58
	STGCN	14.95	23.48	9.87	15.92	25.36	10.42	17.65	28.03	11.34	15.98	25.37	10.43
	GWNet	13.72	21.71	8.80	14.67	23.50	9.49	16.15	25.95	10.74	14.67	23.49	9.52
	ASTGCN	15.72	24.28	10.45	17.85	27.29	11.58	21.96	32.74	14.53	18.17	26.06	11.79
	MTGNN	14.28	22.53	10.54	15.23	24.39	10.52	16.78	26.94	10.88	15.29	24.40	10.68
	GMAN	13.80	22.88	9.41	14.62	24.12	9.57	15.72	26.47	10.56	14.81	24.19	9.69
	STAEformer	13.48	21.91	<u>8.76</u>	14.46	23.86	<u>9.41</u>	15.96	26.63	<u>10.54</u>	14.63	24.13	<u>9.57</u>
	CaST	13.64	21.74	8.79	15.83	25.33	9.86	17.31	29.01	10.88	15.59	25.36	9.84
	TESTAM	<u>13.41</u>	<u>21.79</u>	8.71	<u>14.38</u>	<u>23.73</u>	9.34	<u>15.87</u>	<u>26.48</u>	10.45	<u>14.55</u>	<u>24.00</u>	9.50
	Orion	12.33	18.28	9.83	13.50	20.50	10.79	15.18	23.37	11.42	13.29	20.24	10.55

4.1 Main Experimental Results

Traffic Forecasting Performance. Table 1 reports results on three benchmarks. On METR-LA, Orion attains MAE 2.66 (6.60% gain over STAEformer) with improvements across all horizons. On PEMS04, Orion delivers 8.65% average MAE improvement. On PEMS08, Orion achieves the strongest short-horizon result (8.83% MAE improvement at Horizon-3 over TESTAM) but degrades at longer horizons. We attribute this to PEMS04’s limited temporal coverage (2 months) and its largest node count (307 nodes): fewer periodic variations make the multi-period architecture and directed dependency learning prone to overfitting. This contrasts with METR-LA (4 months, 207 nodes) and PEMS08 (2 months, 170 nodes), where sufficient data diversity or moderate network scale allows Orion to maintain advantages. Short-term gains stem from TE-CausGAT’s ability to capture immediate spatial propagation patterns. Statistical models fail to capture nonlinear spatiotemporal dependencies (e.g.,

METR-LA MAPE > 13%), while recent methods (CaST, TESTAM) narrow the gap. Orion’s directed dependency learning provides consistent short-horizon advantages and competitive long-horizon performance on 2 of 3 datasets.

Epidemic Forecasting Performance. Table 3 shows that Orion generalizes to epidemic prediction, achieving MAE 291.31 on CA-COVID (22.70% improvement over TESTAM) and 30.31 on TX-COVID (37.73% improvement). Unlike traffic networks where spatial relationships are physically constrained by road topology, epidemic spread involves socioeconomic factors and human mobility patterns that cannot be captured by geographical proximity alone. Here, Orion’s prediction-based validation—which integrates prior spatial structure with learned dependencies—is particularly effective at filtering spurious correlations that arise from confounding factors such as shared policy changes or reporting delays.

Effect of LLM-Enhanced Retrieval. We evaluate LLM retrieval on 5-node subsets of PEMS08 and METR-LA, where the LLM selects

Table 2: Ablation study on PEMS-08. Parentheses show percentage change relative to full Orion.

Component	Horizon 3			Horizon 6			Horizon 12			Average		
	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
<i>Part 1: Architecture Components</i>												
Single Period	12.26(+0.57%)	18.83(+3.01%)	8.37(+14.88%)	13.74(+1.78%)	21.65(+5.61%)	8.90(+17.52%)	16.42(+8.17%)	25.97(+11.11%)	10.21(+10.60%)	13.66(+2.78%)	21.36(+5.53%)	8.94(+15.26%)
w/o TE-CausGAT	12.06(-2.19%)	18.14(-0.77%)	8.93(-9.15%)	13.60(+0.74%)	20.80(+1.46%)	10.04(-6.95%)	15.58(+2.63%)	23.96(+2.52%)	11.18(-2.10%)	13.37(+0.60%)	20.37(+0.64%)	9.85(-6.64%)
w/o Multi-Period Fusion	12.03(-2.43%)	18.39(+0.60%)	7.30(-25.79%)	14.02(+3.85%)	21.71(+5.90%)	8.24(-23.63%)	16.88(+11.20%)	25.62(+9.62%)	10.07(-11.82%)	13.65(+2.71%)	20.87(+3.11%)	8.20(-22.27%)
Single Stage Training	12.98(+5.27%)	19.25(+5.31%)	9.46(-3.76%)	15.85(+17.41%)	23.63(+15.27%)	11.19(+3.71%)	20.17(+32.87%)	29.55(+26.42%)	14.15(+23.91%)	15.73(+18.36%)	23.19(+14.57%)	11.46(+8.63%)
Full Orion	12.33	18.28	9.83	13.50	20.50	10.79	15.18	23.37	11.42	13.29	20.24	10.55
<i>Part 2: TE-CausGAT Components</i>												
Single Scale Conv	12.64(+2.51%)	19.35(+5.85%)	9.49(-3.46%)	14.44(+6.96%)	22.37(+9.12%)	10.82(+0.28%)	17.47(+15.09%)	26.96(+15.35%)	12.65(+10.77%)	14.36(+8.05%)	22.09(+9.17%)	10.75(+1.90%)
w/o Intervention	12.11(-1.81%)	18.72(+2.39%)	8.15(-17.09%)	14.05(+4.05%)	21.61(+5.42%)	10.10(-6.40%)	16.44(+8.33%)	25.17(+7.72%)	11.48(+0.53%)	13.80(+3.82%)	21.32(+5.33%)	9.93(-5.88%)
w/o Prior-Guided Fusion	13.22(+7.22%)	20.02(+9.52%)	8.98(-8.64%)	16.05(+18.89%)	24.67(+20.34%)	10.02(-7.14%)	20.66(+36.10%)	31.50(+34.79%)	12.53(+9.72%)	15.92(+19.79%)	24.38(+20.51%)	9.91(-6.07%)
Full Orion	12.33	18.28	9.83	13.50	20.50	10.79	15.18	23.37	11.48	13.29	20.24	10.55

the most similar historical patterns from the past week and month as tri-period input. Table 4 shows that, although overall improvements are moderate (METR-LA reduces average MAPE by 10.57%, PEMS08 by 3.44%), gains in key scenarios are substantial: METR-LA achieves 19.27% MAPE reduction at Horizon-3, and PEMS08 reaches 8.87% at Horizon-12 with a peak of 6.93% at Horizon-6. RMSE shows minimal change (-0.15%) due to sensitivity to extreme values, indicating enhancements primarily benefit typical patterns rather than outliers.

Table 3: Epidemic forecasting performance (lower is better).

Dataset	DGCRN	STAEformer	CaST	TESTAM	Orion
CA	392.56	378.93	396.21	376.84	291.31
TX	58.42	49.36	61.15	48.67	30.31

Table 4: Effect of fine-tuned LLM-enhanced data retrieval (MAPE, %). Hardware: NVIDIA RTX 5880 Ada (48GB) GPU.

Horizon	Metric	PEMS08			METR-LA		
		w/o LLM	w/ LLM	Improv.	w/o LLM	w/ LLM	Improv.
3	MAE	18.77	18.54	1.22%	2.43	2.26	7.00%
	RMSE	32.54	32.05	1.50%	5.21	5.24	-0.71%
	MAPE	9.80	9.64	1.63%	8.20	6.62	19.27%
6	MAE	21.02	20.64	1.82%	3.46	3.31	4.37%
	RMSE	37.35	36.82	1.43%	8.38	8.37	0.10%
	MAPE	11.40	10.61	6.93%	12.41	11.04	11.04%
12	MAE	25.50	25.10	1.59%	4.93	4.83	2.03%
	RMSE	42.47	42.32	0.35%	11.48	11.49	-0.05%
	MAPE	13.76	12.54	8.87%	18.24	17.35	4.88%
Avg.	MAE	21.21	20.86	1.64%	3.47	3.32	4.26%
	RMSE	36.93	36.46	1.28%	8.48	8.49	-0.15%
	MAPE	11.06	10.68	3.44%	12.49	11.17	10.57%

4.2 Ablation Study And Directed Dependency Validation

Ablation study. Table 2 shows that single-stage training causes the largest degradation (18.36% MAE), confirming progressive training is essential. Removing prior-guided fusion yields the largest

single-component loss (19.79% MAE), underscoring integrating priors with learned dependencies. The TE-CausGAT ablation reveals a horizon-dependent pattern: the variant without TE-CausGAT shows marginal MAE reduction at Horizon-3 (-2.19%) but progressive degradation at Horizon-6 (+0.74%) and Horizon-12 (+2.63%). This reflects spatial propagation—short-horizon prediction is dominated by temporal autocorrelation, whereas longer horizons require modeling how perturbations propagate across the network. This holds for multi-period removal (+2.78% average, +8.17% at Horizon-12) and single-scale convolution (+8.05% average, +15.09% at Horizon-12), confirming spatial and temporal components grow more important with horizon. Intervention validation removal increases MAE by 3.82%. Though moderate, this component serves a structural role: it ensures learned edges are predictively reliable, as evidenced by DREAM-3 AUC gain (0.6295 vs. 0.5915 best baseline). The MAE-MAPE divergence reflects metric sensitivity: MAPE is disproportionately affected by near-zero values, while MAE captures accuracy across all magnitudes.

Directed dependency validation on DREAM-3. To quantitatively evaluate Orion’s capability in discovering directed dependencies, we conduct experiments on the DREAM-3 gene regulatory network dataset [29], which provides ground-truth causal structure for benchmarking. As shown in Table 5, Orion achieves an AUC of 0.6295, outperforming the strongest baseline CUTS (0.5915) by 6.42%. Traditional approaches such as PCMC (0.5517) and NGC (0.5579) are constrained by linear assumptions and fixed temporal windows, while neural methods such as eSRU (0.5587) and NGM (0.5477) lack validation mechanisms, often mistaking correlations for directed influence, a gap that Orion’s intervention-based testing directly addresses. The results confirm that our learned structures closely match the ground-truth graph, effectively identifying predictively relevant dependencies that largely align with true causal relationships.

4.3 Case Study: Real-World Application on METR-LA

We validate Orion on METR-LA, whose sensors span major highways (I-405, I-10, US-101). All analyses use the directed dependency matrix averaged over test batches.

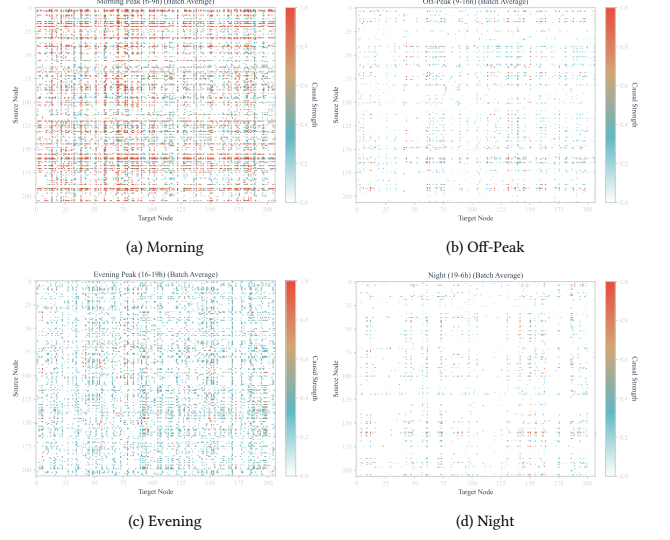
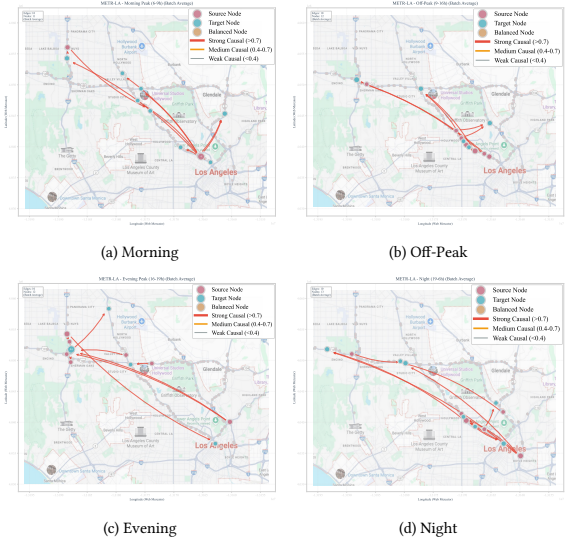
Table 5: Directed dependency discovery on DREAM-3 (AUC; higher is better). Ground-truth structure used for evaluation.

Method	PCMC	NGC	eSRU	LCCM	NGM	CUTS	Orion
AUC	0.5517	0.5579	0.5587	0.5046	0.5477	0.5915	0.6295

Table 6: Key source/target nodes across four periods. Infl.: source out-degree. Src Cnt: target in-degree. Dep. Dist.: top-10 edge count by strength ($S > 0.7$, $M: 0.4-0.7$, $W < 0.4$).

Time Period	Src Node	Infl.	Tgt Node	Src Cnt	Causal Dist.
Morning (6-9h)	Node 4	20	Node 79	2	Strong: 10
	Node 3	3	Node 14	2	Medium: 0
	Node 8	2	Node 22	1	Weak: 0 (Total: 10)
Off-Peak (9-16h)	Node 159	1	Node 139	2	Strong: 6
	Node 160	2	Node 131	3	Medium: 0
	Node 4	2	Node 61	2	Weak: 4 (Total: 10)
Evening (16-19h)	Node 84	1	Node 84	5	Strong: 8
	Node 14	2	Node 91	1	Medium: 0
	Node 29	2	Node 77	1	Weak: 2 (Total: 10)
Night (19-6h)	Node 174	3	Node 73	2	Strong: 0
	Node 160	3	Node 150	2	Medium: 0
	Node 56	1	Node 91	2	Weak: 10 (Total: 10)

Spatiotemporal Evolution of Directed Dependencies. A central advantage of Orion is that its learned directed dependency matrices are time-varying, enabling interpretable analysis of how inter-node influence structures evolve with traffic demand. Figure 6 visualizes these matrices as heatmaps for four representative periods, with edge counts dropping from 12,427 during the morning peak (6–9 AM) to 2,078 at night (7 PM–6 AM). Figure 7 and Table 6 further detail the network topology and key node roles at each period. The morning peak exhibits a “one-to-many” tree-like topology: Node 4 (I-10/I-405 interchange) reaches an out-degree of 20, with influence extending up to 3.5 miles along northbound I-405 and eastbound US-101; 65% of edges connect adjacent sensors while 35% are long-range edges corresponding to detour routes and ramp cascades. The evening peak reverses into a “many-to-one” convergence pattern (9,123 edges), where Node 84 receives strong influences from five upstream nodes (9, 14, 29, 88, 176), forming a bottleneck on I-405. During off-peak hours, dependency density drops by 76.9% and long-range edges decline to 5%, yielding a sparse multi-center structure. At night, only Nodes 174 and 160 remain as independent influence centers, with dependency strengths up to 0.9988. These patterns align with known Los Angeles commuting dynamics, demonstrating that Orion captures physically meaningful, time-varying directed influence rather than static symmetric correlations. Crucially, by identifying *where* influence originates and *how* it propagates, these structures directly inform traffic management: Node 4’s high morning out-degree motivates upstream ramp metering below 40 mph; Node 84’s evening convergence from five sources suggests diverting 30% of traffic when ≥ 3 sources congest; the nighttime dual-center pattern supports dedicated freight lanes at Node 174 and extended green lights at Node 160; and the directed chain between Nodes 14–91 reveals cascading risk addressable by intermediate speed limits.

**Figure 6: Directed dependency matrices on METR-LA across four time periods. Each cell (i, j) denotes dependency strength from node i to node j ; red = strong, blue = weak. Note the progressive sparsification from (a) to (d).****Figure 7: Geographic projection of directed dependencies. Node color: blue (source), red (target), green (balanced). Edge color: red (> 0.7), orange ($0.4-0.7$), yellow (< 0.4). Compare “one-to-many” in (a) with “many-to-one” in (c).**

5 Related Work

From statistical models to deep learning. Spatiotemporal forecasting has progressed from statistical methods to neural architectures. Early approaches such as ARIMA [5] and VAR [18] captured linear dependencies. Deep learning advanced the field: ConvLSTM [36] combined CNNs and LSTMs for spatiotemporal feature extraction, Transformers [39] captured global temporal dependencies, and

iTransformer [28] demonstrated that inverted variate-dimension attention yields strong performance. Large language models have been repurposed for temporal reasoning: Time-LLM [16] repurposes frozen LLMs via prompt-based input transformation, and Time-FFM [27] leveraged LLMs for temporal understanding. Graph neural networks learned node representations through message passing but remained limited to single-dimensional optimization. A shared limitation is these methods learn symmetric correlations without modeling dependency directionality.

Spatiotemporal coupling architectures. To jointly model spatial and temporal dynamics, STGCN [44] introduced graph-temporal convolution, DCRNN combined diffusion convolution with GRU, Graph-WaveNet introduced adaptive adjacency learning, ASTGCN [12] employed spatiotemporal attention, and DGCRN attempted dynamic graph learning. D2STGNN [35] decoupled diffusion and inherent traffic signals, DSTAGNN [20] constructed dynamic spatial-temporal aware graphs, and PDFormer [15] modeled propagation delay effects. In the epidemic domain, MSGNN [32] proposed multi-scale graph convolutions to capture cross-granularity spreading patterns. Despite these advances, three limitations persist: (1) fixed or slowly adapting adjacency matrices cannot represent time-varying connectivity; (2) symmetry assumptions fail to distinguish influence sources from targets; (3) single-scale temporal modeling struggles to capture multi-period patterns.

Causal-inspired spatiotemporal methods. Recognizing these limitations, researchers integrated causal reasoning into forecasting. Assaad et al. [1] categorize causal discovery into constraint-based, score-based, and Granger-based families. PCMC [33] relied on conditional independence testing, NGC [38] and eSRU [17] learned Granger causality [11], and LCCM [9] explored causal connections. DYNOTEARS [31] extended structure learning to time-series under acyclicity constraints, inspiring our DAG regularization. Recently, CaST improves forecasting through learned causal graphs, TESTAM uses spatiotemporal attention for causal discovery, and in the epidemic domain, EpiGNN [47] incorporates epidemiological knowledge into GNNs for heterogeneity-aware forecasting, while Han et al. [13] unify physics- and data-driven modeling via a causal STGNN. However, two limitations remain: static graph assumptions—e.g., CaST uses a single matrix; and lack of validation—learned structures may capture spurious correlations rather than directed dependencies. Leveraging Granger causality and neural causal discovery, we use prediction-based validation to extract predictive edges, avoiding assumptions of structural causal models.

6 Conclusion

We propose **Orion**, a spatiotemporal forecasting framework that learns time-varying directed dependency structures via Granger-inspired estimation and validates them through prediction-based consistency testing, a mechanism that provides a training signal beyond correlation by testing whether learned edge weights predict downstream sensitivity. Combined with multi-scale temporal modeling (Belt Block) and progressive training, Orion achieves state-of-the-art results across five traffic and epidemic benchmarks. DREAM-3 and METR-LA analyses confirm that learned structures capture relevant and physically interpretable dependencies.

A Datasets

Table 7 summarizes the five datasets. Traffic data are high-frequency with rich features, while epidemic data have daily granularity and smaller sample sizes; this diversity enables thorough evaluation of Orion’s generalization across temporal scales and domains.

B Implementation Details

All datasets share consistent base configurations: input/output length 12, four daily periods (6-9 morning peak, 9-16 daytime, 16-19 evening peak, 19-6 nighttime), dropout 0.1 [37], weight decay $1e-4$, and learning-rate decay 0.8 every 20 epochs. The three-stage schedule comprises 10 epochs of feature learning ($\lambda_{causal}=0$), 10 epochs of directed-structure learning ($\lambda_{causal}=0.05$, embedding frozen), and 80 epochs of end-to-end optimization ($\lambda_{causal}=0.03$); GRU propagation is enabled to capture temporal dynamics, all random seeds are fixed at 42, and experiments run on NVIDIA A100-40GB GPUs (traffic) and RTX 2080Ti (epidemic). Dataset-specific values in Table 8 adapt model capacity, learning rate, and intervention frequency to network scale and sample size: the small-sample COVID datasets (86/30 train/test) use $d_{model}=32$ with a single Belt Block and intervention frequency 2 to mitigate overfitting, whereas PEMS04’s 307-node network adopts the smallest learning rate (0.0001) to avoid gradient explosion.

For intervention-based DAG validation, at each batch we sample K nodes via $\text{torch.randperm}(N)[:K]$ ($K=30$ for traffic, $K=25$ for epidemic and DREAM-3), independently of model state and time period. Each selected node i receives an intervention value $v_i = \tanh(g_i(c_i))$, where g_i is a node-specific learnable function and $c_i \sim \mathcal{N}(0, 1)$. During GRU propagation, the intervened node’s input is set to v_i and its incoming edges truncated, mimicking the do-operator. The effect $\Delta \hat{y}_j$ (predictions before vs. after intervention) is then compared with C_{ij} to obtain the validity score (Equation 5); the whole process is differentiable and trained end-to-end.

C LLM Fine-tuning Details

We fine-tune TinyLlama-1.1B-Chat-v1.0 on $\sim 100,000$ time-series-prediction pairs from PEMS03. Each sample pairs 240 historical 5-minute steps (one month) with 12-step predictions, matching Orion’s tri-period window (hourly / daily / weekly), serialized as a two-turn ChatML conversation (user: 240-step sequence; assistant: 12-step targets). Values are min-max normalized to $[0, 1]$ with 4-decimal precision; sequences exceeding the 2048-token context are backward-truncated to retain recent observations. Similarity uses weights $w_{cos} = 0.7$ and $w_{euc} = 0.3$ for semantic matching.

Training uses AdamW (learning rate 2×10^{-5} , weight decay 0.01, 200-step linear warmup then constant) with effective batch size 32 (per-GPU $16 \times$ accumulation 2) for 5 epochs, with gradient checkpointing [7] and mixed-precision training [30] on a single RTX 5880 Ada-48GB GPU. Validation perplexity decreases monotonically from 8.3 to 2.1 and stabilizes after epoch 3 without overfitting, after which the model reliably recognizes recurring traffic motifs (e.g., “morning peak-off-peak-evening peak”) across calendar days, supplying semantic embeddings for the retrieval module. The complete ChatML schema, hyperparameter table, and convergence analysis appear in our supplementary appendix, Section S.4.

Table 7: Dataset statistics

Dataset	Steps	Nodes	Features	Feature Names	Interval	Time Span	Value Range	Unit
PEMS04	16,992	307	3	Flow	5min	Jan-Feb 2018	[0, 919]	veh/h
				Occupancy			[0, 0.772]	ratio
				Speed			[3, 85.2]	mph
PEMS08	17,856	170	3	Flow	5min	Jul-Aug 2016	[0, 1147]	veh/h
				Occupancy			[0, 0.896]	ratio
				Speed			[3, 82.3]	mph
METR-LA	34,272	207	1	Speed	5min	Mar-Jun 2012	[0, 70]	mph
CA-COVID	335	55	3	Cases	Daily	Feb-Dec 2020	[0, 7928]	count
				Population			[0, 7602]	normalized
				Change Rate			[-2263, 2446]	count/day
TX-COVID	335	251	3	Cases	Daily	Feb-Dec 2020	[0, 2149]	count
				Population			[0, 2106]	normalized
				Change Rate			[-569, 381]	count/day

Table 8: Differentiated hyperparameter configurations for main experiments

Parameter	PEMS08	PEMS04	METR-LA	CA-COVID	TX-COVID
Data Configuration					
num_of_vertices	170	307	207	55	251
in_channels	3	3	1	3	3
train_ratio/val_ratio	0.6/0.2	0.6/0.2	0.7/0.1	0.8/0.1	0.8/0.1
num_weeks/days/hours	1/1/1	1/1/1	1/1/1	0/0/1	0/0/1
Model Configuration					
d_model	64	64	64	32	32
d_ff*	128	128	128	64	64
n_belt_block	2	2	2	1	1
head_t/head_f	4/4	4/4	4/4	2/2	2/2
Training Configuration					
batch_size	128	64	96	16	16
learning_rate	0.005	0.0001	0.0005	0.005	0.001
Intervention Configuration					
intervention_frequency	10	10	10	2	2
num_interventions	30	30	30	25	25

D Reproducibility, Limitations, and Ethics

Reproducibility. To facilitate independent verification, we publicly release all experiment code, configuration files, and data preprocessing scripts at the anonymous repository linked in Section 1. Our implementation is built on PyTorch 2.0+ with experiments conducted on NVIDIA A100-40GB GPUs (traffic datasets), RTX 2080Ti-22GB GPUs (epidemic datasets), and RTX 5880 Ada-48GB GPU (LLM retrieval). Deterministic execution is ensured by fixing all random seeds to 42 across Python, NumPy, PyTorch, and CUDA backends. Every hyperparameter is exhaustively documented in the appendix and our supplementary appendix, including three-stage training schedules (Table 8 and our supplementary appendix, Section S.3.3), intervention details (Appendix B), and preprocessing procedures (our supplementary appendix, Section S.6). All five datasets are publicly available benchmarks with standard community splits. The repository includes a comprehensive README.md

with step-by-step instructions for environment setup, data preparation, and end-to-end training.

Limitations. Although Orion achieves strong results across five benchmarks, several practical constraints should be noted. The intervention-based validation requires extra forward passes during training, adding roughly 15–20% wall-clock overhead compared to a correlation-only baseline; in practice this is kept manageable through the intervention frequency hyperparameter and by restricting the causal learning phase to 10 of the 100 total epochs, with no additional cost at inference time. On PEMS04, which covers only 2 months with 307 nodes, the multi-period architecture tends to overfit at longer horizons—a data scarcity issue rather than a modeling flaw, as the same design yields consistent gains on METR-LA (4 months) and PEMS08 (170 nodes). Our polynomial DAG constraint penalizes only 2- and 3-cycles and thus cannot theoretically rule out longer cycles, though empirical verification via topological sorting detects none on PEMS-08 at threshold 0.1 (our supplementary appendix, Section S.2). The TinyLlama retrieval module has $O(N \times H)$ complexity, and due to limited GPU memory availability, full-network LLM-based retrieval remains computationally prohibitive for datasets with hundreds of nodes, restricting current evaluation to 5-node subsets obtained via K-means; all primary results reported in Table 1 rely on fixed-period retrieval and are therefore unaffected by this constraint.

Ethics. This study uses exclusively publicly available datasets containing no personally identifiable information. The traffic datasets (PEMS04, PEMS08, METR-LA) consist of aggregated loop-detector readings recording only vehicle counts, occupancy ratios, and speeds; the COVID-19 datasets (CA-COVID, TX-COVID) provide county-level hospitalization statistics from Dong et al. [10]; and the DREAM-3 dataset [29] comprises entirely synthetic gene expression trajectories. All usage complies with respective distribution licenses and privacy regulations. Orion is designed as a decision-support tool whose forecasts and dependency structures are intended to inform—rather than replace—human judgment in traffic management and public health planning. No human subjects were involved at any stage of this research, and no demographic or sensitive attributes are modeled or inferred.

GenAI Usage Disclosure

We disclose all use of generative AI (GenAI) tools in this work as follows.

As a methodological component. Our framework integrates a fine-tuned TinyLlama-1.1B-Chat-v1.0 model as part of the LLM-enhanced retrieval module described in Section 3.1. The LLM is used solely for computing semantic similarity between temporal segments and is fine-tuned offline once on the PEMS03 dataset. It is frozen during Orion’s training (loaded with `model.eval()` and queried under `torch.no_grad()`) and serves only as an inference-time retrieval component; it does not generate any scientific content of the paper.

For writing assistance. GenAI tools (large language models) were used to polish English grammar and improve clarity of expression in selected sections of the manuscript. All research ideas, problem formulation, methodology design, theoretical analysis, experimental design, and result interpretation were conceived and completed solely by the authors. No GenAI tool generated novel scientific content, derived theoretical results, or designed experiments.

For code and data. GenAI assistants were used in a limited capacity to help with routine implementation tasks (e.g., LaTeX formatting and code refactoring suggestions). All algorithmic logic, model implementation, and experimental pipelines were authored, reviewed, and validated by the authors. No GenAI tool was used to generate, modify, or synthesize experimental data.

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